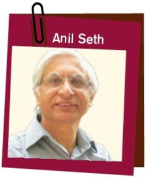

Visualising Data Using googleVis

Viewing statistics in motion charts can be great fun. This article describes how googleVis software can be used to view motion charts. The author gives the reader an interesting example of data picked up from the Bombay Stock Exchange and how it can be viewed as a motion chart.

Viewing statistics can be an exciting experience. If you are surprised, look at one of the motion chart presentations by Hans Rosling, at his TED talk *https://www.youtube.com/watch?v=RUwS1uAdUcI.* The software behind the presentation of motion charts was acquired by Google and integrated into the Google Charts API.

Normally, I would have experimented with the Python API. However, I came across the R package 'googleVis' on GitHub and decided to experiment with that instead. It provides a motivation to experiment with R.

R is a statistical tool, which assumes considerable significance in the context of making sense of Big Data.

Installing googleVis

You need to install R and then the googleVis package under R. For example, on Fedora, use the following commands:

```
$ sudo dnf install R
$ sudo R
>install.packages("googleVis")
```

You may be prompted to select a repository from which to install R packages. Once the installation is complete, you can test it with just two commands, as follows:

```
$ R
> suppressPackageStartupMessages(library(googleVis))
> plot(gvisMotionChart(Fruits, "Fruit", "Year",
options=list(width=600, height=400)))
starting httpd help server ... done
```

You will get a 600x400 (width x height) motion chart in the browser and can play around with various bubbles moving with time. You can find videos of googleVis charts on YouTube in case you are interested.

Getting started with R

In case you are not already familiar with R, at this point, try a sample session in 'An Introduction to R' which is included in the R distribution. You can access the built-in help including the introduction as follows:

```
> help.start()
starting httpd help server ... done
```

This will bring up HTML documentation. You can navigate to 'Appendix A: A sample session' in 'An Introduction to R'.

Exploring googleVis

Now, let's get back to understanding and exploring the googleVis package.

'Fruits' is a data frame included in the distribution of the googleVis package. The command *data()* will list all the data frames available. However, we are currently interested in the googleVis package only. So, find out the data frames available in googleVis, load *Fruits* and print its contents as follows:

```
> data(package="googleVis")
Data sets in package 'googleVis':
Andrew                  Hurricane Andrew: googleVis example
                        data set
Cairo                   Daily temperature data for Cairo
CityPopularity          CityPopularity: googleVis example
                        data set
Exports                 Exports: googleVis example data set
Fruits                  Fruits: googleVis example data set
….
> data(Fruits)
> Fruits
    Fruit Year Location Sales Expenses Profit       Date
1  Apples 2008     West    98       78     20 2008-12-31
2  Apples 2009     West   111       79     32 2009-12-31
...
9 Bananas 2010     East    81       71     10 2010-12-31
```